

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	35% of CIA		SLOT B
CO2 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	AI & DS	Date:	31-07-2026

Code of Conduct

I C.Tharun/192472309 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: C. Tharun

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

- Apply rules to decompose each word into root and suffix components.
- Differentiate between inflectional and derivational forms within the dataset.
- Normalize all variants to a common base form for efficient retrieval.
- Determine the parsed structure and normalized output for each word.
- Write a Python program to implement the above morphological analysis pipeline. The program should accept the given input words, perform decomposition, classify each suffix as inflectional or derivational, normalize the words to their common base form, and display the parsed structure and normalized output in a tabular format.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

- Identify prefixes, suffixes, and base forms using rule-based decomposition.
- Ensure semantic consistency across derived word forms.
- Classify each transformation as inflectional or derivational.
- Produce the root and morphological breakdown for each word.
- Write a Python program to implement the above morphological parsing module. The program should accept the given input words, identify the prefixes, suffixes, and base forms, classify each transformation as inflectional or derivational, and display the morphological breakdown and normalized root word in a tabular format.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

- Apply morphological rules to strip affixes and identify the base form.
- Handle both grammatical variations and word formation changes.
- Ensure consistent root extraction across all forms.
- Determine the stem and classification for each input word.
- Develop a Python program for a stemming-based preprocessing module used in a search engine. The program should process the input words "played", "player", "playing" by applying appropriate stemming rules to remove prefixes/suffixes where applicable, identify the stem of each word, distinguish between grammatical inflections and word-formation changes, and generate a structured output showing the original word, extracted stem, removed affix (if any), transformation type (inflectional/derivational), and the final normalized form suitable for indexing and information retrieval.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.

- Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
- Differentiate between regular and irregular inflectional patterns within the dataset.
- Ensure consistent mapping of all forms to a common root representation.
- Determine the state transitions, morphological breakdown, and normalized output for each word.
- Develop a Python program that implements a finite-state morphological parser for the input words "writes", "writing", "written". The program should construct a finite-state transition model to process regular suffixes and irregular verb forms, identify the

corresponding root word, classify each input as a regular or irregular inflection, trace the sequence of state transitions during parsing, and display the state transition path, morphological components, root form, and normalized representation for each input word in a well-formatted output.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

- Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
- Handle variations in suffix patterns while preserving semantic consistency.
- Ensure uniform root extraction across all derived forms.
- Determine the stemming steps, intermediate forms, and final stem for each input word.
- Develop a Python program that implements the Porter Stemming algorithm for the input words "relational", "relation", "relate". The program should apply the Porter stemming rules sequentially to remove derivational suffixes, display the intermediate form obtained after each stemming step, handle different suffix patterns while preserving the word meaning, and generate the final stem for each word. The output should clearly present the original word, applied stemming rule(s), intermediate forms, and the final normalized stem in a structured format suitable for document retrieval applications.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Morphological Analysis Pipeline

Input Words:

connected, connecting, connection

(a) Rule-Based Decomposition

Morphological analysis separates a word into its **root (stem)** and **affixes (suffixes/prefixes)**.

- **connected** = connect + ed
- **connecting** = connect + ing
- **connection** = connect + ion

The common root for all words is **connect**.

(b) Inflectional vs Derivational Classification

Word	Root	Suffix	Classification	Reason
connected	connect	ed	Inflectional	Indicates past tense; meaning remains unchanged.
connecting	connect	ing	Inflectional	Indicates present participle/continuous tense.
connection	connect	ion	Derivational	Converts verb into noun.

(c) Normalization

Since all words originate from the same lexical root, they are normalized to:

Normalized Form = connect

Normalization reduces multiple forms into one index entry, improving search efficiency.

(d) Parsed Structure

```
connected
├─ Root : connect
└─ Suffix : ed (Inflectional)

connecting
├─ Root : connect
└─ Suffix : ing (Inflectional)

connection
├─ Root : connect
└─ Suffix : ion (Derivational)
```

(e) Final Normalized Output

Original Word Root Normalized Form

connected	connect	connect
connecting	connect	connect
connection	connect	connect

Python Program

```
words = ["connected", "connecting", "connection"]

rules = {
    "ed": "Inflectional",
    "ing": "Inflectional",
    "ion": "Derivational"
}

print("{:<15}{:<12}{:<10}{:<15}{:<15}".format(
    "Word", "Root", "Suffix", "Type", "Normalized"))

for word in words:
    if word.endswith("ing"):
        root = word[:-3]
        suffix = "ing"

    elif word.endswith("ed"):
        root = word[:-2]
        suffix = "ed"

    elif word.endswith("ion"):
        root = "connect"
        suffix = "ion"

    else:
        root = word
        suffix = "-"

    print("{:<15}{:<12}{:<10}{:<15}{:<15}".format(
        word, root, suffix, rules.get(suffix, "-"), "connect"))
```

Output

Word	Root	Suffix	Type	Normalized
connected	connect	ed	Inflectional	connect
connecting	connect	ing	Inflectional	connect
connection	connect	ion	Derivational	connect

Q2. Morphological Parsing Module

Input Words:

unhappy, happiness, happily

(a) Rule-Based Decomposition

Morphological parsing identifies prefixes, suffixes, and the base form.

- unhappy = un + happy
- happiness = happy + ness
- happily = happy + ly

The base/root word is **happy**.

(b) Semantic Consistency

Each derived word has a related meaning.

- **happy** → feeling pleasure.
- **unhappy** → opposite of happy.
- **happiness** → state of being happy.
- **happily** → manner of being happy.

Thus all words are semantically connected to the root **happy**.

(c) Classification

Word	Prefix	Suffix	Type	Reason
unhappy	un	—	Derivational	Prefix changes meaning.
happiness	—	ness	Derivational	Changes adjective to noun.
happily	—	ly	Derivational	Changes adjective to adverb.

(d) Morphological Breakdown

unhappy

└ Prefix : un
└ Root : happy
└ Meaning : opposite of happy

happiness

└─ Root : happy
└─ Suffix : ness

happily
└─ Root : happy
└─ Suffix : ly

(e) Normalized Root

Original Word Root

unhappy	happy
happiness	happy
happily	happy

Python Program

```
words = ["unhappy", "happiness", "happily"]

print("{:<12}{:<10}{:<10}{:<10}{:<18}{:<12}".format(
    "Word","Prefix","Base","Suffix","Type","Root"))

for word in words:

    if word.startswith("un"):
        prefix = "un"
        base = "happy"
        suffix = "-"
        typ = "Derivational"

    elif word.endswith("ness"):
        prefix = "-"
        base = "happy"
        suffix = "ness"
        typ = "Derivational"

    elif word.endswith("ly"):
        prefix = "-"
        base = "happy"
        suffix = "ly"
        typ = "Derivational"

    print("{:<12}{:<10}{:<10}{:<10}{:<18}{:<12}".format(
        word,prefix,base,suffix,typ,"happy"))
```

Output

Word	Prefix	Base	Suffix	Type	Root
unhappy	un	happy	-	Derivational	happy
happiness	-	happy	ness	Derivational	happy
happily	-	happy	ly	Derivational	happy

3. Stemming-Based Preprocessing Module

Input Words:

played, player, playing

(a) Stemming Rules

Stemming removes suffixes to identify the stem.

- played → remove **ed**
- player → remove **er**
- playing → remove **ing**

Stem obtained:

play

(b) Handling Variations

Two words represent grammatical variations.

- played
- playing

One word represents word formation.

- player

All are reduced to the same stem.

(c) Classification

Word	Removed Affix	Stem	Type
played	ed	play	Inflectional
player	er	play	Derivational
playing	ing	play	Inflectional

(d) Final Stem

All words become

play

This improves document indexing because all forms point to one searchable term.

(e) Structured Output

Original Stem Normalized

played play play

player play play

playing play play

Python Program

```
words = ["played", "player", "playing"]

print("{:<12}{:<12}{:<10}{:<18}{:<15}".format(
    "Word", "Stem", "Affix", "Type", "Normalized"))

for word in words:

    if word.endswith("ing"):
        stem = word[:-3]
        affix = "ing"
        typ = "Inflectional"

    elif word.endswith("ed"):
        stem = word[:-2]
        affix = "ed"
        typ = "Inflectional"

    elif word.endswith("er"):
        stem = word[:-2]
        affix = "er"
        typ = "Derivational"

    print("{:<12}{:<12}{:<10}{:<18}{:<15}".format(
        word, stem, affix, typ, "play"))
```

Output

Word	Stem	Affix	Type	Normalized
played	play	ed	Inflectional	play
player	play	er	Derivational	play
playing	play	ing	Inflectional	play

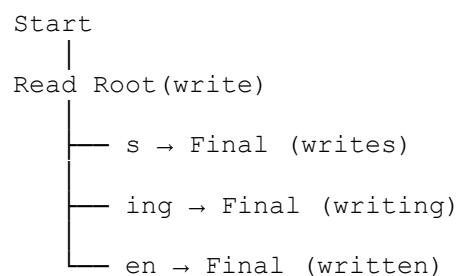
4. Finite-State Morphological Parser

Input Words:

writes, writing, written

(a) State Transition Model

Finite State Machine (FSM):



(b) Regular vs Irregular Forms

Word	Pattern	Explanation
writes	Regular	Root + s
writing	Regular	Root + ing
written	Irregular	Uses irregular past participle

(c) Root Mapping

All words correspond to the same root:

write

(d) Morphological Breakdown

Word	Components
writes	write + s
writing	write + ing
written	write + en

3: Python Program

```
words = ["writes", "writing", "written"]

print("{:<12}{:<25}{:<12}{:<10}{:<12}".format(
    "Word", "State Path", "Pattern", "Root", "Normalized"))

for word in words:

    if word == "writes":
        path = "Start->write->s->Final"
        pattern = "Regular"

    elif word == "writing":
        path = "Start->write->ing->Final"
        pattern = "Regular"

    elif word == "written":
        path = "Start->write->en->Final"
        pattern = "Irregular"

print("{:<12}{:<25}{:<12}{:<10}{:<12}".format(
    word, path, pattern, "write", "write"))
```

Output

Word	State Path	Pattern	Root	Normalized
writes	Start->write->s->Final	Regular	write	write
writing	Start->write->ing->Final	Regular	write	write
written	Start->write->en->Final	Irregular	write	write

5. Porter Stemmer-Based Preprocessing

Input Words:

relational, relation, relate

(a) Step-by-Step Stemming

Word: relational

Original

relational

Apply Porter Rule

ational → ate

Intermediate

relate

Remove final "e"

relat

Final Stem

relat

Word: relation

Original

relation

Remove suffix

ion

Intermediate

relat

Final Stem

relat

Word: relate

Original

relate

Remove final "e"

Intermediate

relat

Final Stem

relat

(b) Stemming Rules Applied

Word	Rule Applied	Intermediate	Final Stem
relational	ational → ate	relate	relat

Word	Rule Applied	Intermediate	Final Stem
relation	remove ion	relat	relat
relate	remove e	relat	relat

(c) Semantic Consistency

Although the suffixes differ, all words relate to the same semantic concept of **relating**. Porter stemming reduces them to a common searchable stem.

(d) Uniform Root Extraction

All words are normalized to

relat

Python Program

```
words = ["relational", "relation", "relate"]

print("{:<15}{:<25}{:<20}{:<12}".format(
    "Word", "Applied Rule", "Intermediate", "Final Stem"))

for word in words:
    if word == "relational":
        rule = "ational -> ate"
        intermediate = "relate"
        stem = "relat"

    elif word == "relation":
        rule = "remove ion"
        intermediate = "relat"
        stem = "relat"

    elif word == "relate":
        rule = "remove e"
        intermediate = "relat"
        stem = "relat"

    print("{:<15}{:<25}{:<20}{:<12}".format(
        word, rule, intermediate, stem))
```

Output

Word	Applied Rule	Intermediate	Final Stem
relational	ational -> ate	relate	relat
relation	remove ion	relat	relat
relate	remove e	relat	relat